

INFO 4335

Mobile Application Development

FINAL GROUP PROJECT — INSTRUCTION & ASSESSMENT GUIDE

Assessment Weight	Group Size	Project Due	Final Demo
35% of Final Grade	Assigned by Lecturer	18 June 2026	19 June 2026

1. Project Overview

This final assessment (35%) requires each group to propose, design, develop, and present a fully functional hybrid mobile application using Flutter. The project must demonstrate mastery of the INFO 4335 Course Learning Outcome 3 (CLO 3): working as a team to develop a functional mobile application, while exhibiting effective communication with diverse stakeholders (PLO 5).

CLO 3: Work as a team to develop a functional mobile application.

PLO 5: Exhibit effective communication with diverse stakeholders. Students must demonstrate clear, professional communication throughout the proposal, report, and live demonstration.

⚠ Individual Work: This project must be completed as a group. However, every team member is expected to contribute actively and meaningfully to the work. Each member's level of contribution will be assessed through a **5-minute individual video presentation**, where members are required to present their own contributions to the project.

2. Application Domain

Select a topic from (but not limited to) the following Shariah-compliant domains:

Domain	Examples
Healthcare	Appointment booking, medication tracker, mental wellness
Education	Quiz app, e-learning, student portal
Islamic Da'wah / Community	Prayer times, Quran reader, event management
Business / F&B	Order management, inventory, food delivery
Sports & Fitness	Workout planner, nutrition log, team management

Domain	Examples
Tourism	Travel guide, itinerary planner, local discovery
Social Welfare & NGO	Donation tracker, volunteer management, fundraising
Productivity / Utilities	Task manager, notes, time tracker

Shariah-Compliant Requirement: All topics, content, images, and features must adhere to Islamic ethical standards. No inappropriate content, advertising, or features are permitted.

3. Part 1 — Project Proposal

Due Date: 5 June 2026

Submission: Submit the GitHub repository URL via iTa'leem. The proposal must be written in GitHub README.md (Markdown format). The proposal carries **no marks**. It is submitted solely to assess whether your project scope is sufficient and aligned with the course guidelines.

⚠ Important: A missing, incomplete, or outdated README.md will be treated as NO submission. Ensure the README reflects your latest work at all times.

3.1 Mandatory README.md Sections

Your proposal README.md must include ALL of the following sections:

#	Section	Required Content
1	Group Members	Full names and matric numbers of all group members; clearly state each member's assigned role
2	Project Title	Clear, descriptive title for your application
3	Introduction	Problem statement, motivation, and relevance of the proposed app
4	Objectives	Specific, measurable objectives the app aims to accomplish
5	Target Users	Description of the primary and secondary user groups
6	Features & Functionalities	Core modules, interactions, and UI components with brief descriptions
7	UI Mock-up	Sketches or wireframes of at least 5 key screens
8	Architecture / Technical Design	Widget/component structure and chosen state management approach (e.g., setState, Provider, Riverpod, Bloc)
9	Data Model	ERD or Firestore collection-document model diagram
10	Flowchart / Sequence Diagram	User interaction and navigation flow across the app
11	References	Flutter docs, Firebase docs, UI design references, and any other sources

4. Part 2 — Development Requirements (35%)

Part 2 comprises three components: Coding (10%), Final Report (15%), and Final Presentation & Demo (10%), totalling 35%.

4.1 Coding Requirements (10%)

Your final application must satisfy all criteria below. Examiner will inspect the source code and run the application live. The raw score is out of 100 marks, then converted to 10%.

4.1.1 Core Development Criteria

- Correct and idiomatic use of Flutter widgets/components
- Responsive UI layout using Rows, Columns, Containers, ListView, and GridViews
- Proper use of forms, text fields, validators, and user interactions
- Navigation implemented with named routes or go_router
- State management implemented using a recognised approach (setState, Provider, Riverpod, or Bloc)
- Integration of at least two external packages or plugins from pub.dev

4.1.2 Firebase Integration (At Least Two (2) Services Required)

Firestore Service	Usage Example
Authentication	Email/password login, Google Sign-In, anonymous auth
Cloud Firestore	Real-time data storage, CRUD operations, queries
Firestore Storage	Image/file upload and retrieval
Cloud Messaging	Push notifications, topic-based messaging

4.1.3 Teamwork & Collaboration Evidence (CLO 3)

As this project is assessed under CLO 3 (teamwork), the following evidence of collaboration is required:

- GitHub commit history must show regular, meaningful contributions from ALL members
- Each member must be assigned and own specific modules or features
- README.md must include a clear task division table listing each member's responsibilities
- Each member will be individually assessed based on their own contribution to the project.

4.1.4 Software Engineering Practices

- Clean, well-organised project folder structure
- Modular and reusable widget/component design
- Meaningful naming conventions (classes, variables, files)
- Code is free of critical errors and warnings
- Compliant with Flutter best practices and coding conventions

4.1.5 UI/UX Design Standards

- Consistent and visually coherent colour scheme throughout the app
- Appropriate use of typography, iconography, and spacing
- User-friendly navigation and interaction patterns
- Shariah-compliant imagery, content, and messaging

4.2 Coding Marks Breakdown (100 Marks → 10%)

Ref	Assessment Area	Marks
B1	UI Layout & Widgets/Components	15
B2	Navigation & Routing	10
B3	Forms & User Inputs	10
B4	State Management Implementation	15
B5	Use of Packages & Plugins	10
B6	Firebase Integration (Auth / Firestore / Storage)	15
B7	Use of Media (images, audio/video, files)	10
B8	UI/UX Design Quality	10
B9	Code Structure & Best Practices	5
	TOTAL	100 → 10%

4.3 Final Report — in PDF format (15%)

Each team member is required to write and submit their own individual report. Group submissions are **not accepted**. Your report must reflect **your own contribution only**. Do not include work done by other team members. Every student must submit separately via iTa'leem. It must include:

1. All items from **Section 1 to 10** as outlined in the proposal
2. Final UI screenshots of all major screens (**your role/parts only**)
3. Summary of all achieved features mapped to original objectives
4. Technical explanation of architecture, state management approach, and Firebase usage (**your role/parts only**)
5. Reflection on teamwork process: how tasks were divided, challenges faced, how they were resolved
6. Known limitations and proposed future enhancements

4.4 Final Presentation & Demo (10%)

Date: 19 June 2026 | Duration: 5 minutes video demo presentation (no slides required).

Ref	Component	Description	Marks
D1	Demo Quality	App runs without errors; all major features demonstrated live	4
D2	Individual Presentation & Communication	Each member presents their specific module clearly and professionally to the audience	6
	TOTAL		10

Absence Policy: Members who fail to submit their final video presentation without a valid documented reason will be awarded a **ZERO** for this section.

5. Assessment Summary & Grading Scheme

No	Component	Submission Method	Max Marks	Weight
1	Coding & Implementation	GitHub source code via iTa'leem	100 → 10	10%
2	Final Report (PDF)	iTa'leem	15	15%
3	Final Presentation & Demo	Video demo 19 June 2026 via iTa'leem	10	10%
	GRAND TOTAL		35	35%

6. Submission Guidelines

Stage	What to Submit	Where / When
Part 1	GitHub URL — README.md with full proposal, team roles, and mock-ups	iTa'leem — Part 1 link Due: 5 June 2026
Part 2	GitHub URL — all source code	iTa'leem — Part 2 link Due: 18 June 2026
Demo	Live demonstration on device or emulator; all members present	19 June 2026 (Video Presentation) via iTa'leem

GitHub Repository Requirements: Repository must be public or shared with the lecturer. Commit history must show regular, ongoing contributions from ALL group members. A single large final commit will be flagged for academic integrity review.

7. Academic Integrity

- Plagiarism between groups will result in zero marks for all parties and may be escalated to the faculty.
- Proper attribution must be given to any open-source code, libraries, or design assets used.
- The lecturer reserves the right to conduct individual oral questioning to verify each member's contribution. Members unable to explain their own code may have their individual marks adjusted.

8. Recommended Resources

Resource	URL / Description
Flutter Documentation	https://docs.flutter.dev
Firebase Documentation	https://firebase.google.com/docs
pub.dev (Flutter Packages)	https://pub.dev
Flutter State Management	https://docs.flutter.dev/data-and-backend/state-mgmt/options
Material Design 3	https://m3.material.io
GitHub Markdown Guide	https://docs.github.com/en/get-started/writing-on-github

— *End of Group Project Instructions* —

Prepared by: Mohd Khairul Azmi Hassan | INFO 4335 — Mobile Application Development